

LAB 0

TECH 60 - Sec 14

F26

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Objective:

The objective for LAB 0 was to understand the resistor color code and identify the resistance value of a resistor by reading the color bands. We verified the data and checked the answers by utilizing a Digital Multimeter.

Equipment/Components:

Digital Multimeter

Various resistors (3)

Digital Multimeter cables

1. How do resistors play a role in electrical circuits?

They help control the flow of current through the circuit.

2. Explain the significance of color codes and tolerance values applied to resistors.

The color codes and tolerance values provide a standard visual system for identifying the resistance and tolerance limits without actively measuring them with a DMM.

3. Discuss the findings from Table 1 and Table 2. Can a measured resistance value fall within the specified tolerance range? Could it fall outside of the specified range. Explain the reasoning behind your answer.

Yes, the resistance value can fall within and outside the specified tolerance range. Most resistors come with the guaranteed tolerance labeled on them, but due to burnouts and improper handling, they can fall outside the tolerance.

4. Discuss Ohm's Law and how a resistor affects the current in circuit for a given voltage.

Ohm's law is $I = V/R$. The resistance is inversely proportional to the current when given a fixed voltage.

Data:



Table 1:

Color Code Band	1 st Band Color, digit	2 nd Band Color, digit	3 rd Band Color, digit	4 th Band Color, digit	Nominal Value	% Tolerance
Resistor 1	Brown 1	Green 5	Brown x10	Gold	150	5
Resistor 2	Orange 3	Black 0	Brown x10	Gold	300	5
Resistor 3	Brown 1	Brown 1	Brown x10	Gold	110	5

Table 2:

Color Code Band	Nominal Value	Tolerance	Min Value	Max Value	Measured value	Percent Error
Resistor 1	150	5	142.5	157.5	152 ohms	1.3%
Resistor 2	300	5	285.0	315.0	301 ohms	0.3%
Resistor 3	110	5	104.5	115.5	108 ohms	1.8%

Hand Calculations:

Resistor 1: $(15\ \Omega \times 10) \pm 5\% = 150\ \Omega \pm 5\%$

Resistor 2: $(30\ \Omega \times 10) \pm 5\% = 300\ \Omega \pm 5\%$

Resistor 3: $(11\ \Omega \times 10) \pm 5\% = 110\ \Omega \pm 5\%$

Challenges Encountered:

Some challenges that we encountered during this lab include some faults with the equipment. As Table 2 shows, two of our percent errors (R1, R2) were above the 1% threshold. For the purpose of this initial lab, the value is acceptable, but for future labs, the calculations will need to be redone and become more precise. We also had a bit of a hard time finding the right place to put the resistors back because of their size, but by using the color code chart from the lab manual, we were able to put everything in its original spot.

Percent Error Calculations:

Resistor 1: $((150\ \Omega - 152\ \Omega) / 150\ \Omega) \times 100\% = -1.3\%$

Resistor 2: $((300\ \Omega - 301\ \Omega) / 300\ \Omega) \times 100\% = -0.3\%$

Resistor 3: $((110\ \Omega - 108\ \Omega) / 110\ \Omega) \times 100\% = 1.8\%$

Conclusion: So we learned that resistors are needed in a circuit to limit the flow of electrical current and lower voltage levels so sensitive parts do not need a break. The color codes and the tolerance values provide a universal, physical method to identify a component's electrical resistance and accuracy when the part is too small to print legitimate numbers.